UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 2B: Evaluation Methods

EIN 3241: Human Factors and Ergonomics
Spring 2026

January 23, 2026

Dr. Jad Atweh

Admin Issues

- Quiz 1 Monday

Recap

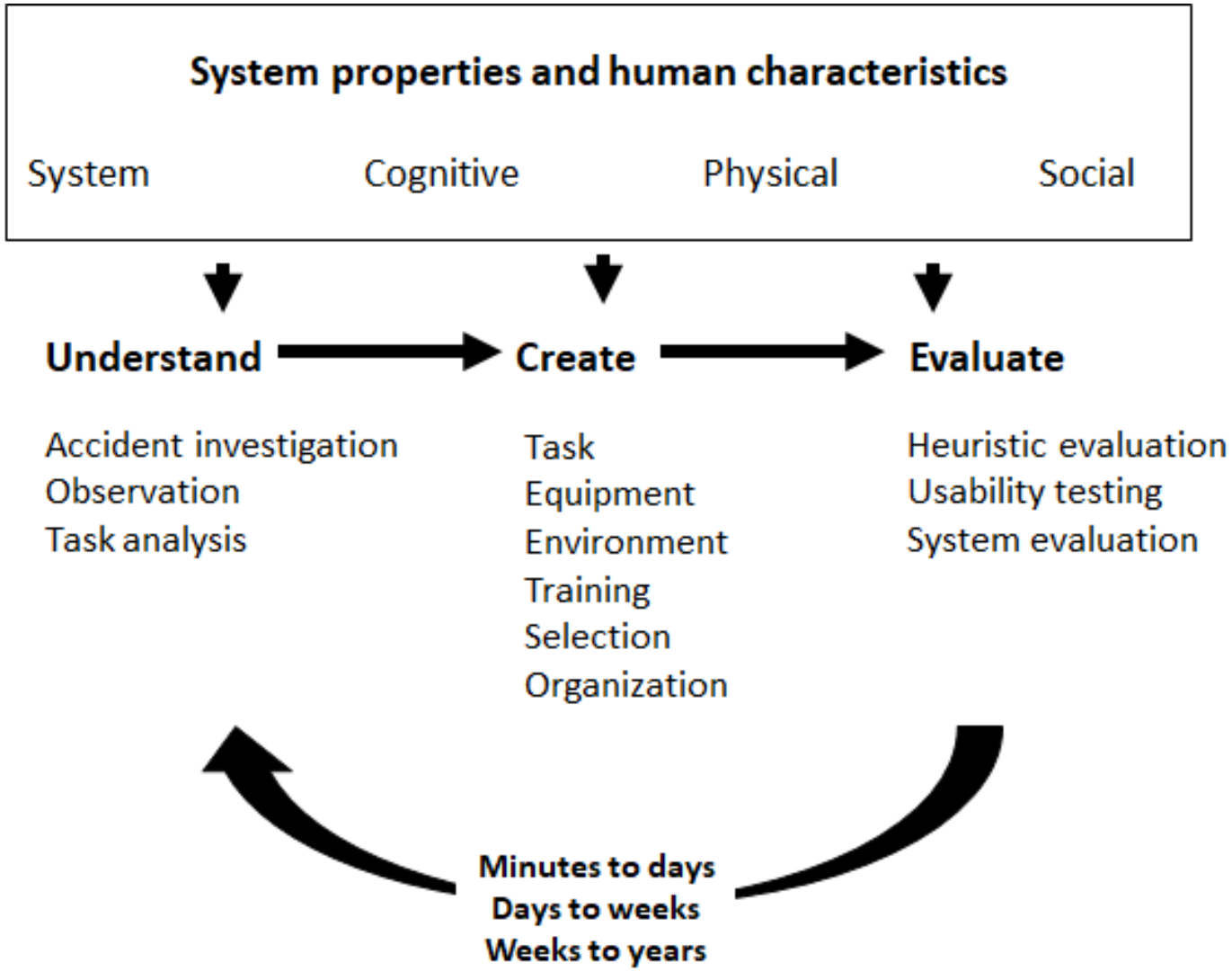
- Task Analysis: What and Why..
- Types of Task Analysis
  - Hierarchical Task Analysis (HTA)
  - Cognitive Task Analysis (CTA)
  - Contextual Task Analysis
  - Sequential Task Analysis
  - Physical Task Analysis
- Dived into HTA, CTA:
  - Plans…
  - GOMS…
- How to Perform a Task Analysis:
  - Define the purpose and identify the data required
  - Collect task data
  - Interpret task data
  - Innovate from task data
    - Personas
    - Scenarios
    - User Journeys
    - Use Cases

Learning Objectives

By the end of this set of lectures, you should be able to:

- Understand the purpose of evaluation in HF/E
- Recognize differences between evaluation methods and how they support the human design cycle
- Understand the role of formative and summative human factors evaluations

Human Factors Design Cycle



Design and Prototyping (Create step)

- Depends on two types of understanding:
  - Understanding tasks (front-end analysis/understand step)
  - Knowledge of general human capabilities

- Design is iterative
  - Prototype fidelity (low, medium, high)
  - Evaluation at each stage

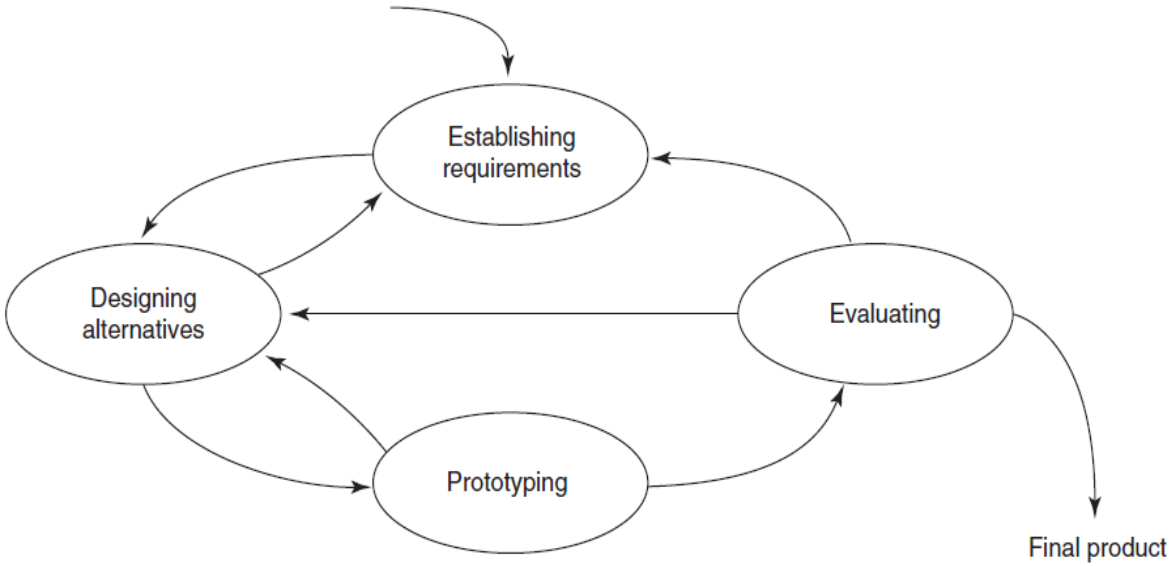
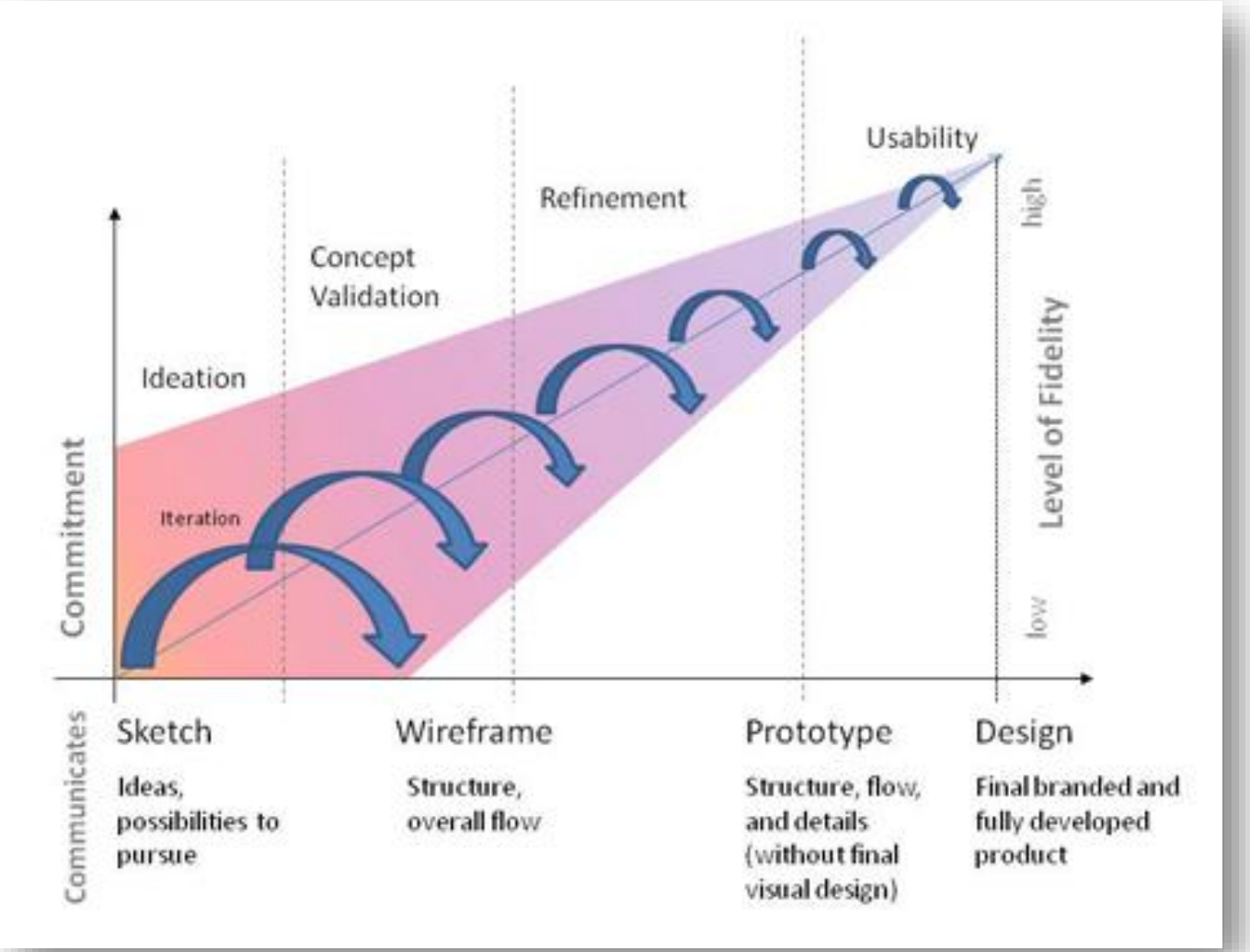


Figure 9.3 A simple interaction design lifecyle model

Prototyping

- Paper prototypes
- Wireframes
- High-fidelity prototypes



https://www.uxmatters.com/mt/archives/2010/05/sketches-and-wireframes-and-prototypes-oh-my-creating-your-own-magical-wizard-experience.php

What is Evidence?

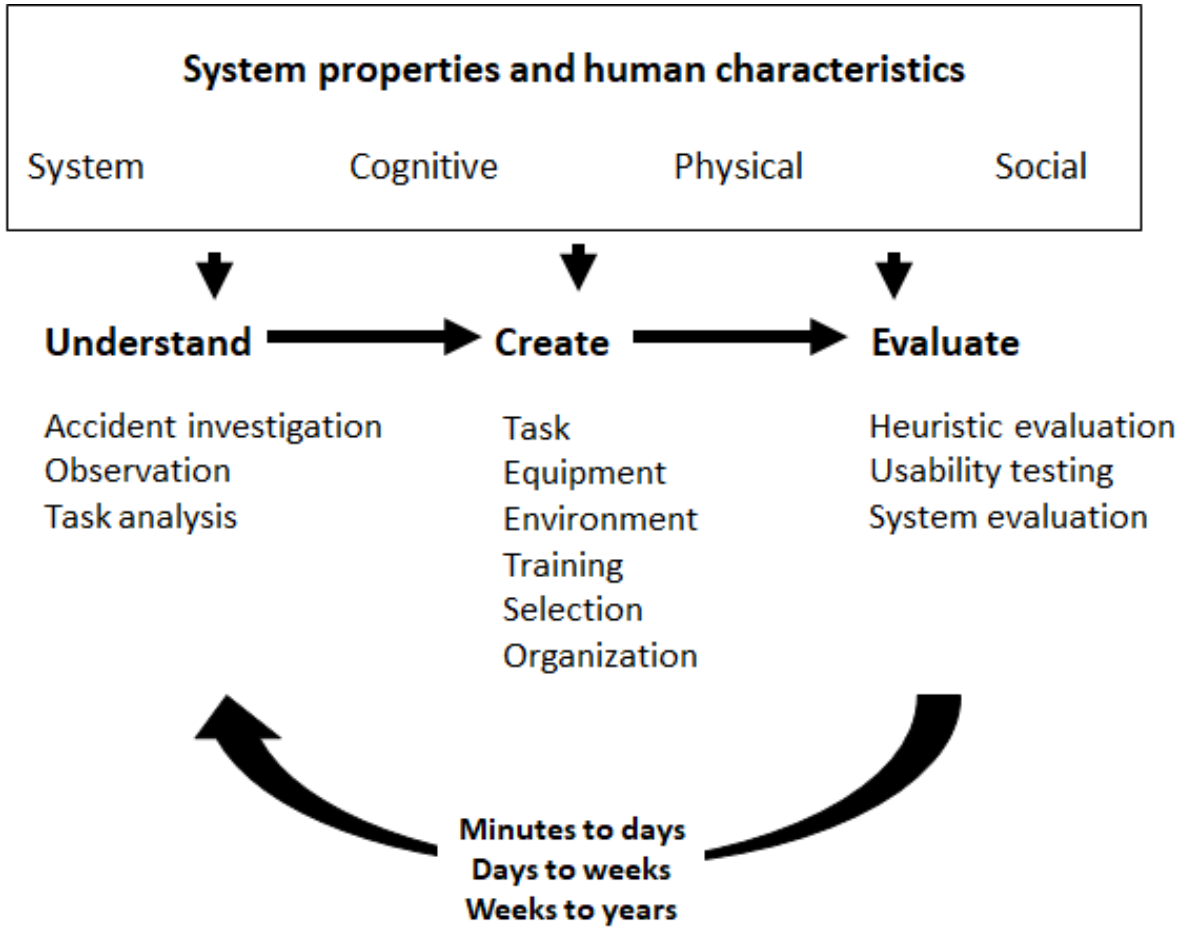
- Now that we looked at design methods.. We need to support it.
- Evidence = Data that supports design decisions
- Can be qualitative (observations, interviews) or quantitative (task time, error rates)
- Helps answer:
  - Does the design meet user needs?
  - Is it usable, safe, and efficient?
  - What should be improved?

Purpose of Evaluation

- To ensure systems are usable, safe, and effective
- To identify and fix design flaws before deployment
- To understand user needs, behavior, and context
- Design evaluation
  - Testing specific prototypes, design solutions, or systems
  - e.g., apple watch vs. in-vehicle infotainment systems
- Concept evaluation
  - Basic science that supports the design principles and heuristics
  - e.g., visual distraction, cognitive distractions, manual distractions

Evaluation in the Human Factors Design Cycle

- Evaluation is iterative and ongoing
- Used to:
  - Discover user needs (early)
  - Test design concepts (mid-stage)
  - Validate final product (late stage)
- Embedded in all phases: research, prototyping, development, deployment

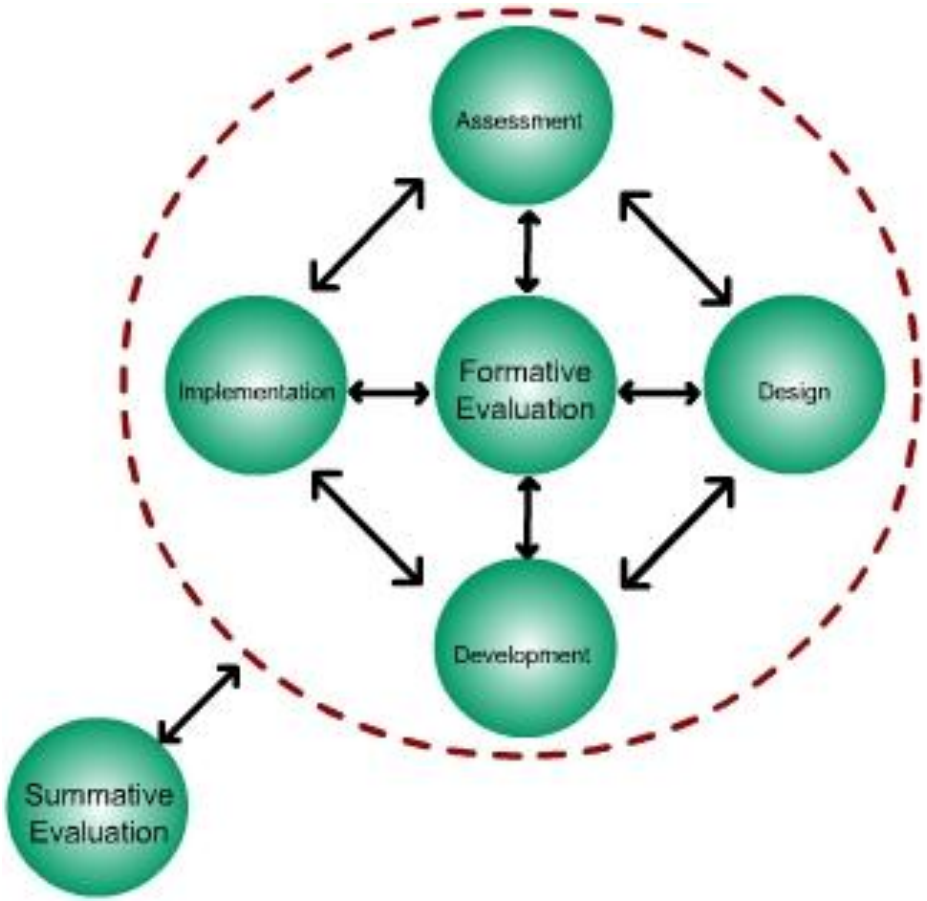


Types of Evaluation: Formative VS Summative

Formative Evaluation:

- Conducted during design
- Helps improve the design
- Involves early feedback
- Diagnose problems with prototypes
- Qualitative measures
- Less formal than summative evaluations
- FDA:

  "Formative evaluations are used to inform device user interface design while it is in development. It should focus on the issues that the preliminary analyses indicated were most likely to involve use safety (e.g., aspects of user interaction with the device that are complicated and need to be explored). It should also focus on those areas where design options for the user interface are not yet final."



FDA Guidelines: *"Applying Human Factors and Usability Engineering to Medical Devices"*

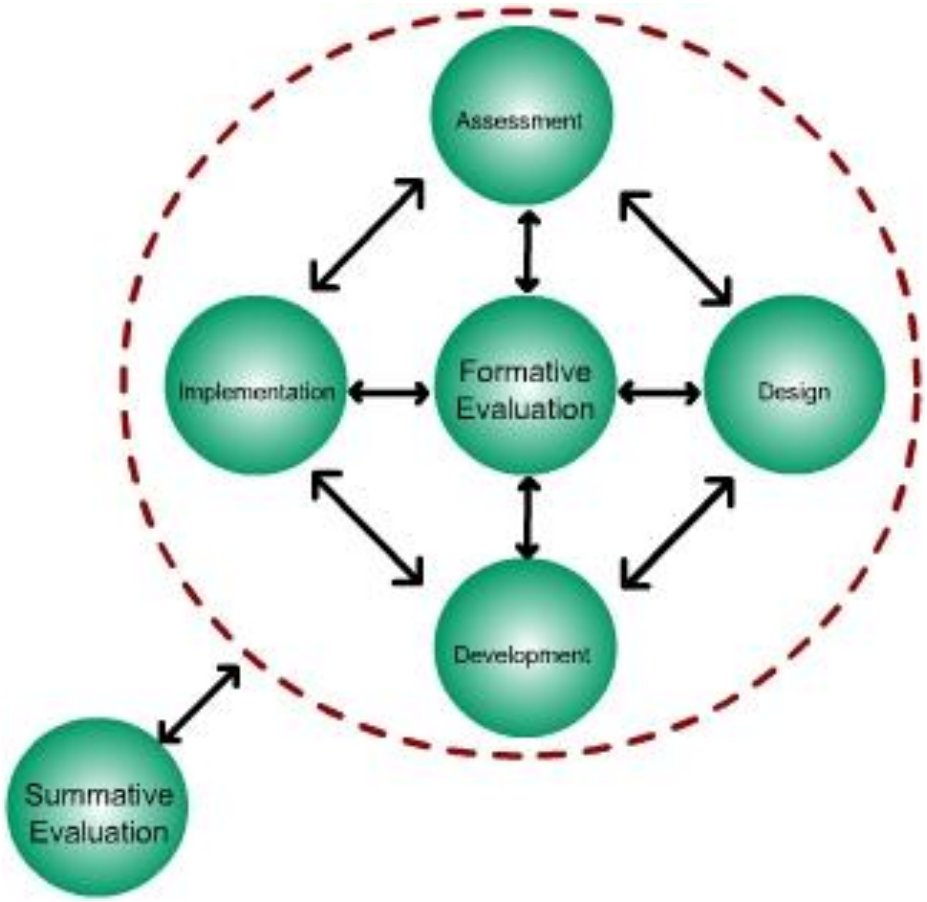
Types of Evaluation: Formative VS Summative

Summative Evaluation:

- Assess whether a design meets benchmarks or design goals
- Conducted after development
- Verify and validate
- Quantitative measures (measure an underlying variable – Annett's Evaluative Methods)
- Measures usability, performance, satisfaction
- Supports decision-making (e.g., go/no-go)
- FDA: Human factors validation testing

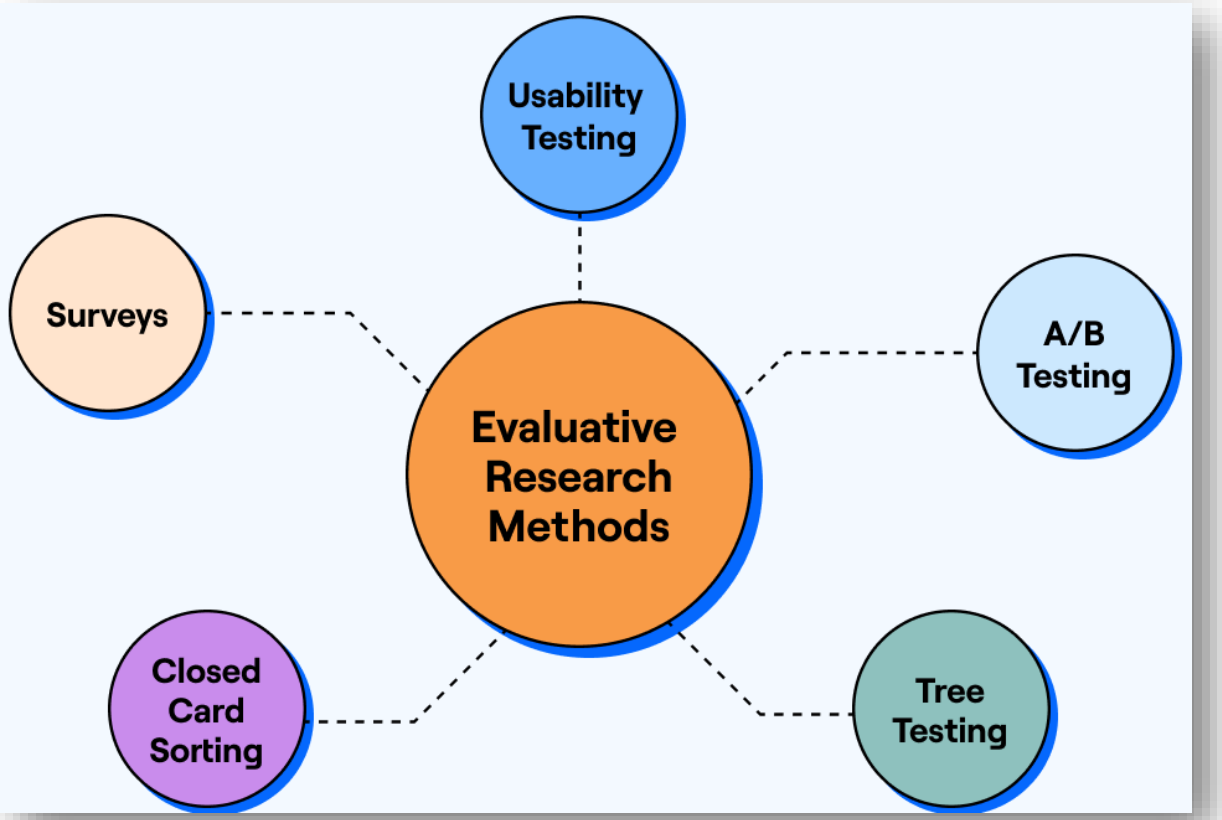
  *"Human factors validation testing is conducted to demonstrate that the device can be used by the intended users **without serious use errors or problems**, for the **intended uses** and under the **expected use conditions**. The testing should be comprehensive in scope, adequately sensitive to capture use **errors** caused by the design of the user interface and should be performed such that the results can be generalized to actual use."*

FDA Guidelines: *"Applying Human Factors and Usability Engineering to Medical Devices"*



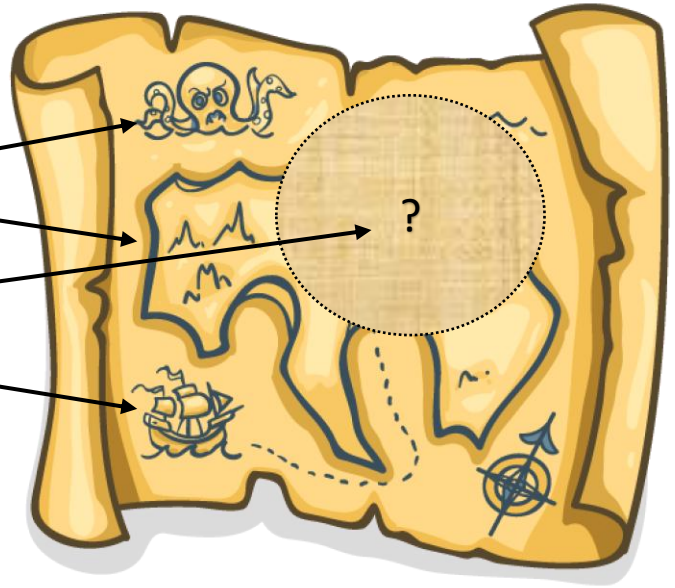
Evaluative Methods

- Literature Reviews
- Heuristic Evaluations
- Cognitive Walkthroughs
- Usability Testing
- In-service evaluation
- A-B Testing
- Surveys & Questionnaires
- Field Studies / Observational Methods
- Controlled Experiments
- ...



Literature Reviews

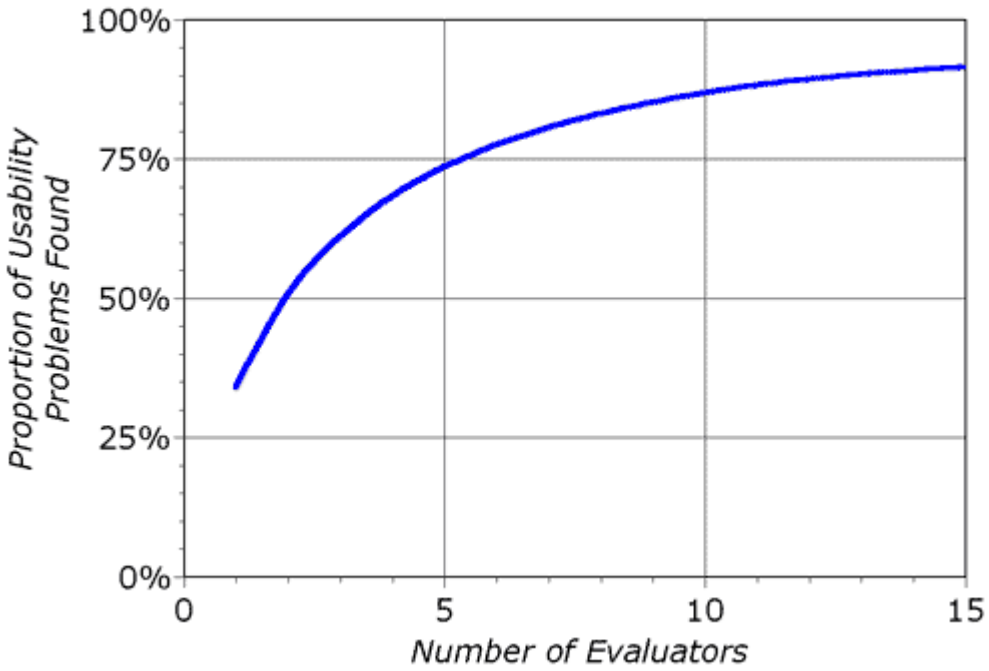
- What is the current state of the art ?
- What is the scholarly debate?
- Where are the gaps?
- How do my results compare?



Lay of the Land

Heuristic Evaluations

- HF/usability expert evaluates a design based on heuristics (guidelines, principles, rules of thumb)
- No additional data collection from users
- Cost effective and fast process
- Best suited for interface design
- 3-5 evaluators typically
- Fast, cost-effective
- Often used early in design
- Limitations: subjective, depends on evaluator expertise



https://www.interaction-design.org/literature/article/heuristic-evaluation-how-to-conduct-a-heuristic-evaluation

Heuristic Evaluation Process

1. Identify and select human factors principles that are particularly applicable.
2. Carefully inspect the design and identify where it violates the design principles.
3. Aggregate findings for each evaluator and communicate to design team.

Nielsen User Interface Design Heuristics

Visibility of System Status

1

Match Between System & Real World

2

User Control And Freedom

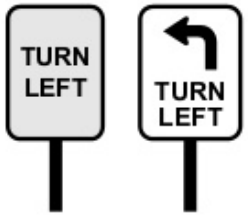
3

Consistency And Standards

4

Error Prevention

5

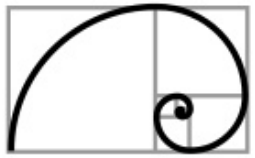


Recognition Rather Than Recall

6

Flexibility And Efficiency of Use

7



Aesthetic And Minimalististic Design

8

Help Users With Errors

9

Help And Documentation

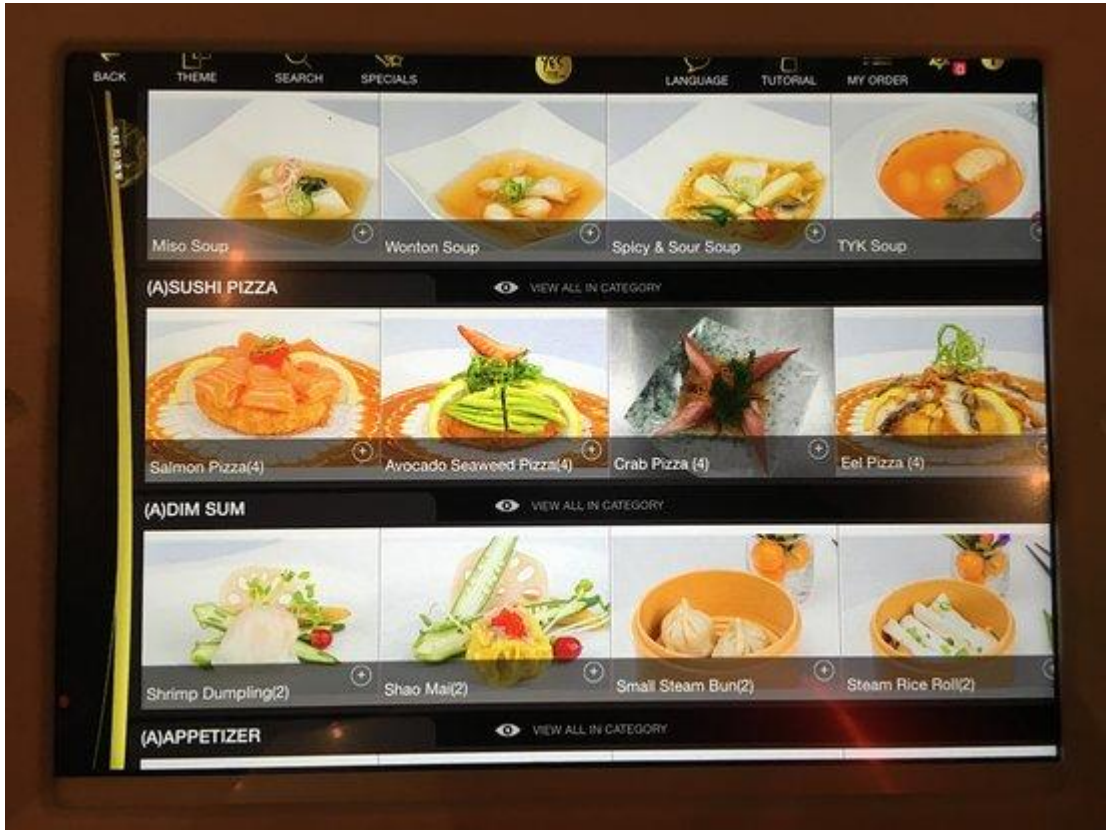
10

Cognitive Walkthrough

- Similar to heuristic evaluation
  - Expert evaluation
  - Highlight potential problems with the design
- Take on perspective of user
  - Step-by-step analysis of how users solve tasks
- Requires a specific set of tasks/use cases
- Best suited for interaction design
- Focus on learnability for new or infrequent users
- Ask: "Will the user know what to do next?"
- Useful for interfaces with complex workflows

Cognitive Walkthrough

- Is it likely that the person will perform the right action?
- Does the person understand what task needs to be performed?
- Will the person notice that the next task can be performed?
- Will the person understand how to perform the task?
- Does the person get feedback after performing the task indicating successful completion?



Evaluative Methods

- Literature Reviews
- Heuristic Evaluations
- Cognitive Walkthroughs

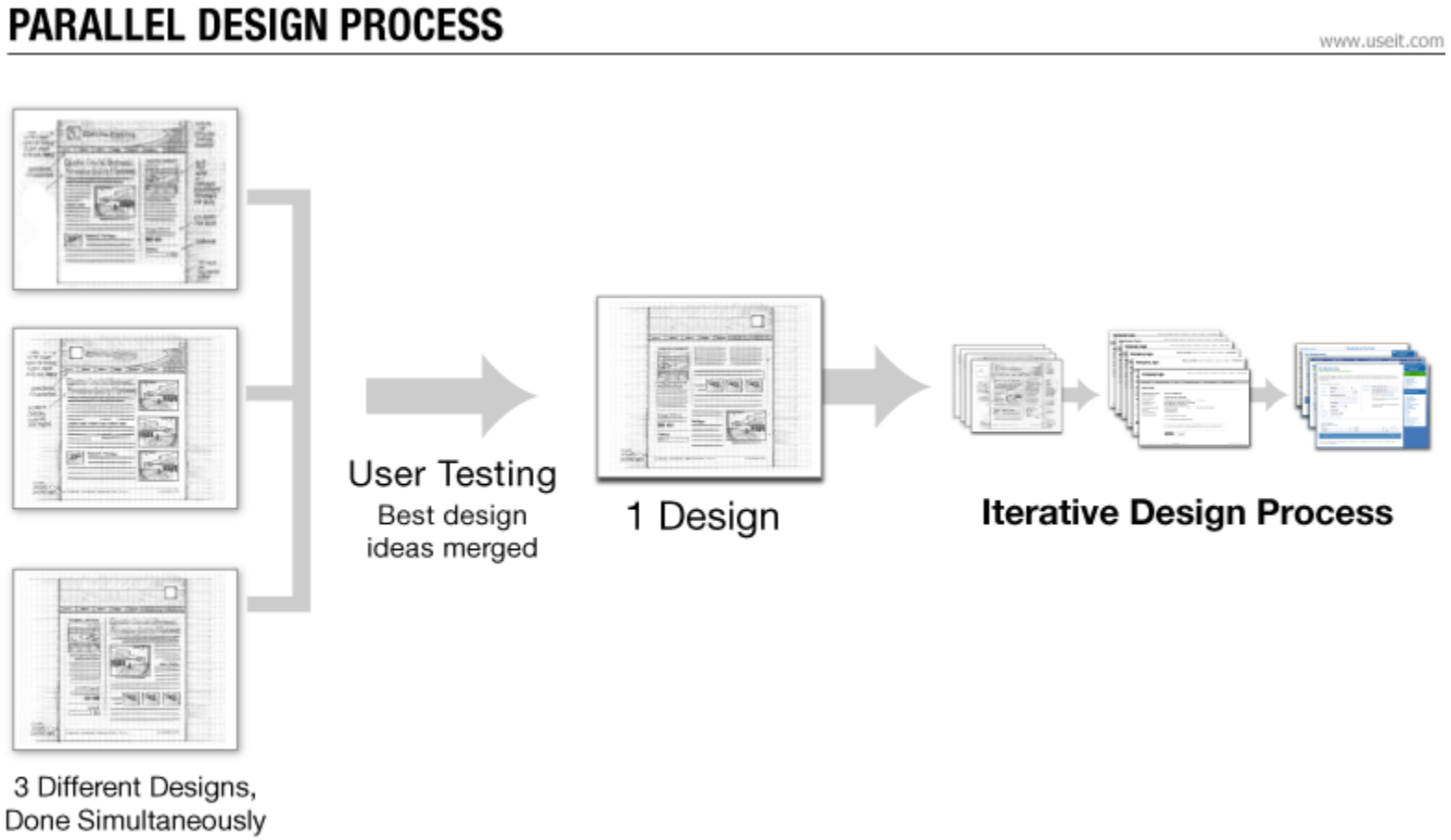
What's common between them?

- Usability Testing
- In-service evaluation
- A-B Testing
- Surveys & Questionnaires
- Field Studies / Observational Methods
- Controlled Experiments
- …

Usability Testing

- Testing a design with representative users
- Focuses on the usability (ease of use) of a design
  - Learnability
  - Efficiency
  - Memorability
  - Errors
  - Satisfaction
- Can be lab-based, remote, or in-field
- Can differ in scope
  - Small n, descriptive statistics, qualitative analysis
  - Large n, inferential statistics, quantitative analysis
- Example: Testing a mobile app prototype

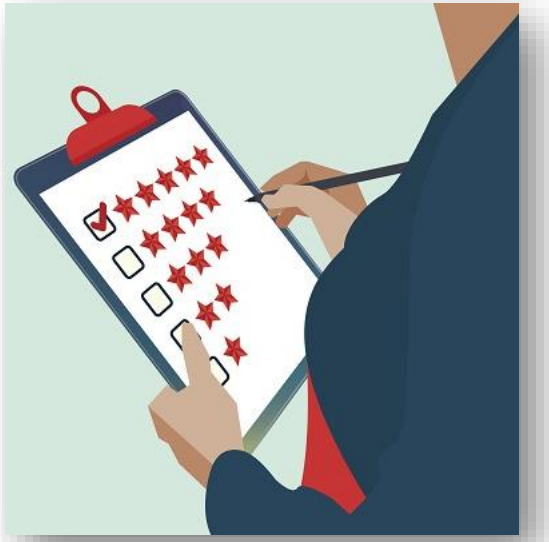
Usability Testing



https://www.nngroup.com/articles/parallel-and-iterative-design/

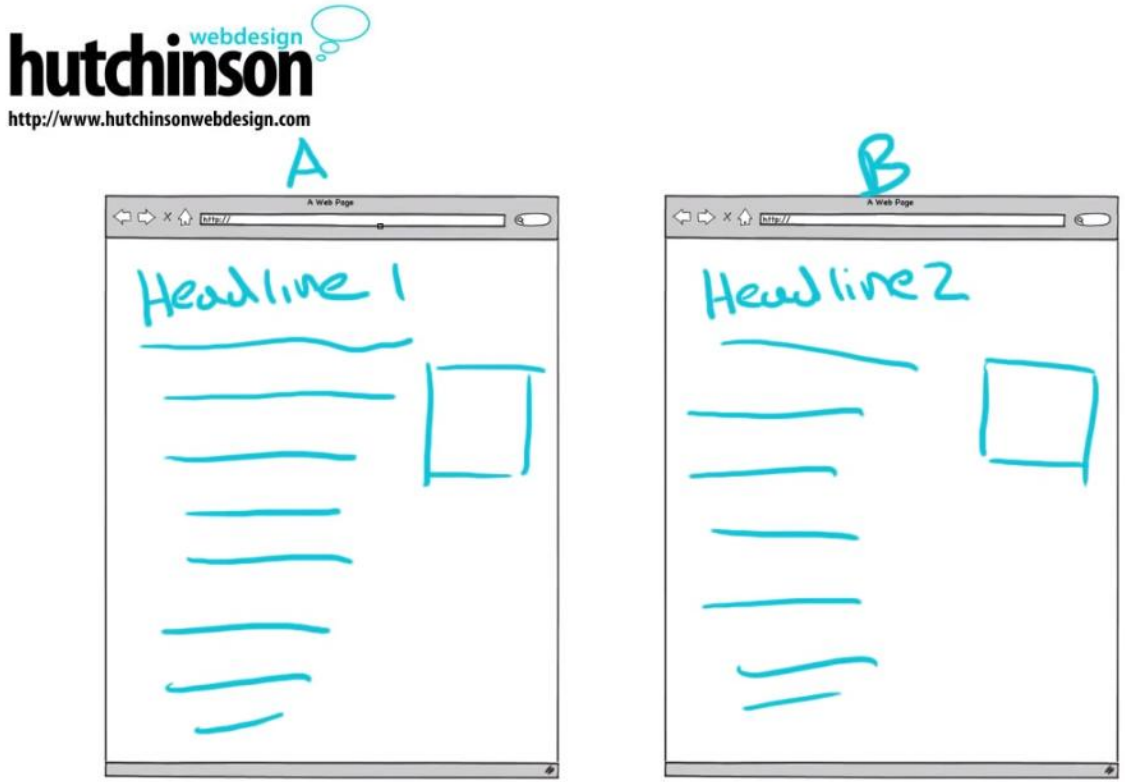
In-Service Evaluation

- In-service evaluation refers to the ongoing assessment of a system, product, or process after it has been deployed and is in actual use — i.e., it's already in service.
- Summative system assessment
- Ideally done within real-world contexts
- Controlled experiments
  - Compare new system versus status quo (e.g., clinical trial)
  - Compare measures of interest against benchmarks
- Examples:
  - Monitoring nurse interaction with electronic health records months after implementation
  - Gathering feedback from pilots on a new interface after flight hours accumulate
  - Evaluating the maintenance logs and downtime of a newly installed machine in a factory



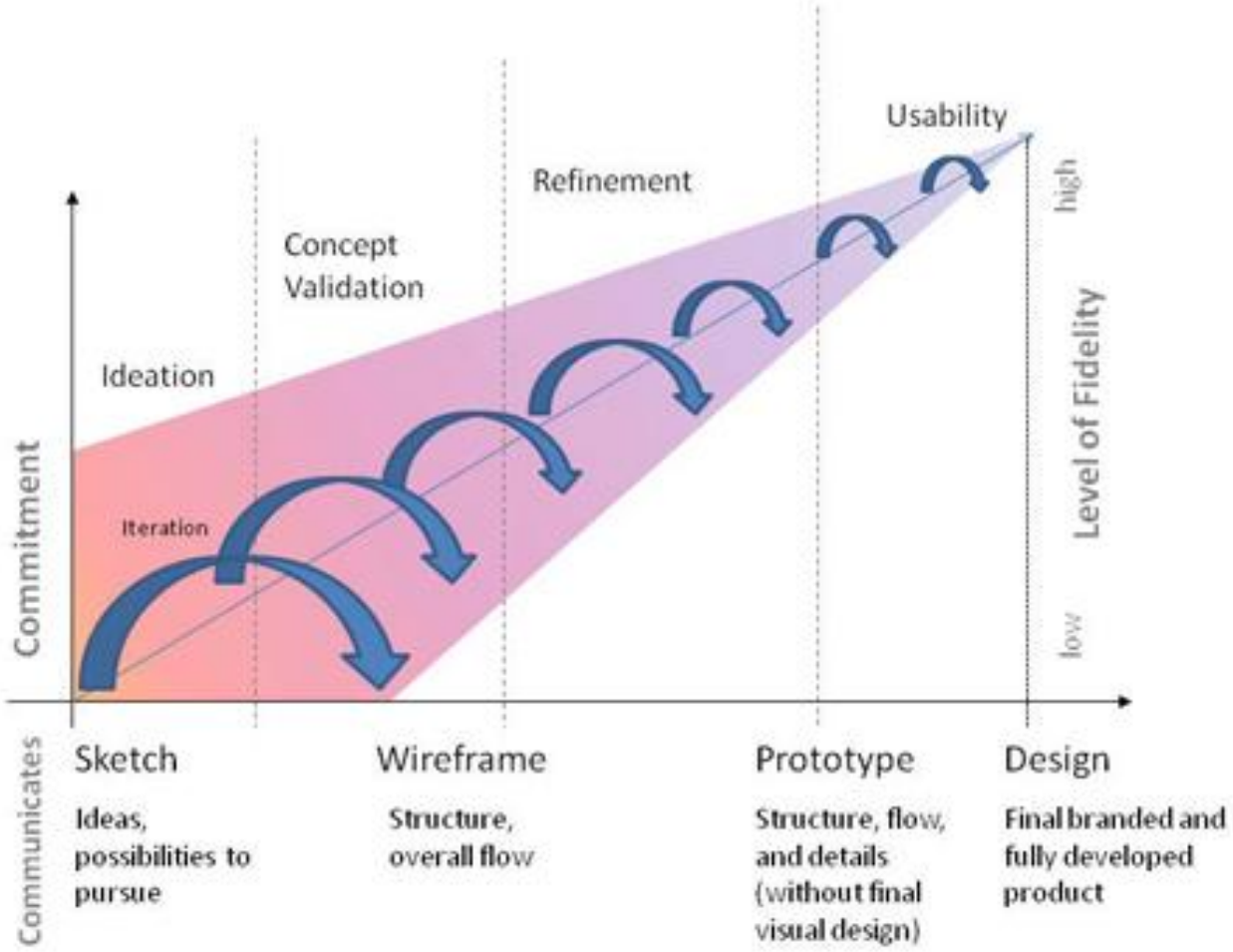
A/B Testing

- Also known as split-testing
- Two systems (or more; e.g., webpages) evaluated against each other
- Small change (e.g., the headline) and test conversic rate (people who sign up or click through).
- Very large number of participants typically
  - Facebook / Google
- Can be used to test very small differences (slightly different shades of blue)
- Best for minor design variations, not big conceptual differences
- Doesn't explain why one version worked better — just which one did



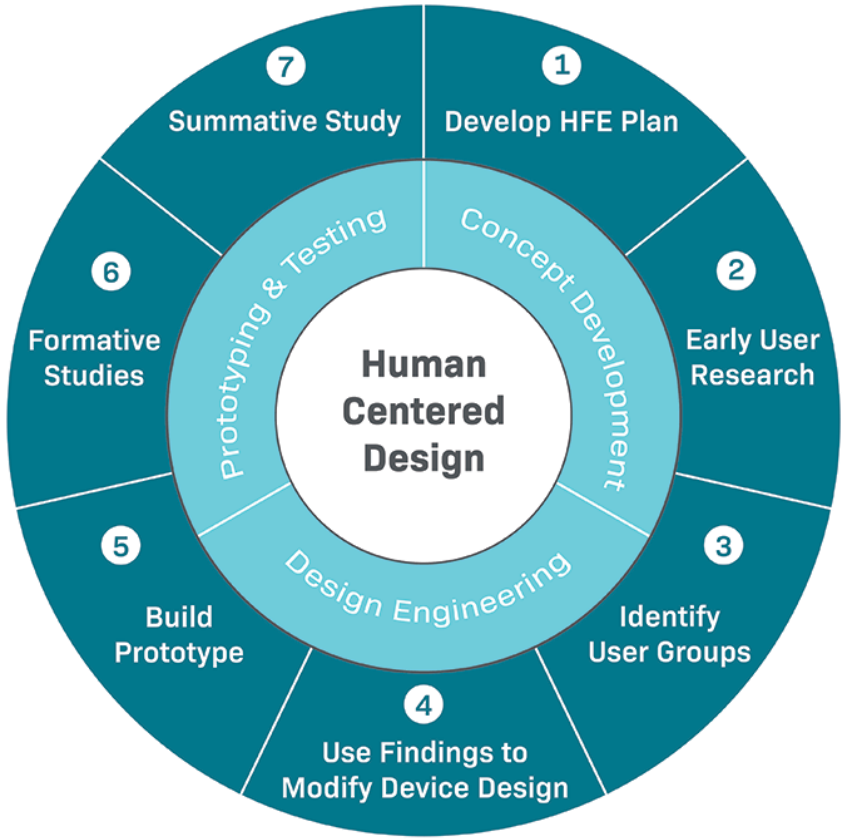
Timing and Types of Evaluation

- Conceptual Design:
  - Literature Review
  - Heuristic evaluation
- Prototyping & Iteration:
  - Cognitive walkthrough
  - Heuristic evaluation
  - Usability test
- Pre-production model:
  - User study
  - A-B Testing
- Post-Deployment:
  - In-service monitoring
  - A-B Testing



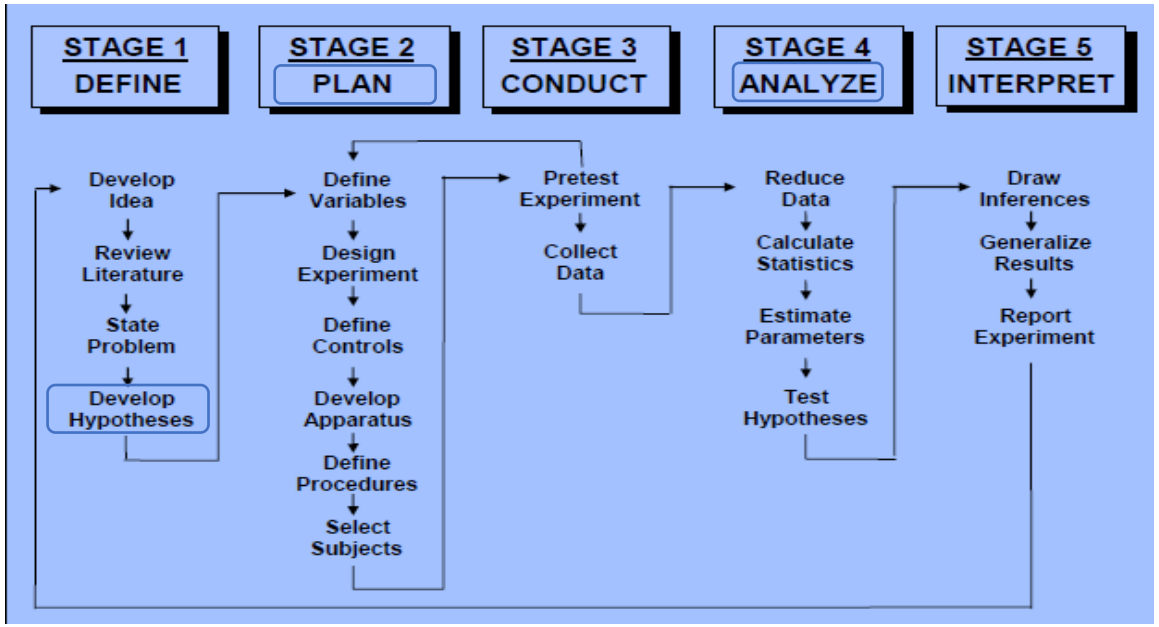
What did we learn today?

- Evaluation is a core part of human-centered design
- Evaluation types: formative and summative. Difference?
- Different Evaluation Methods
- Choose methods that align with your goals and design stage
- Mix qualitative and quantitative data for robust insights
- Evidence from evaluation guides better design decisions



Next time…

- HFE Research Methods
- Stages of Research
- Brush up on Hypothesis testing
- Experimental Design
- Limitations in Research
- Challenges in HFE Research
- …



UF

Herbert Wertheim
College of Engineering
UNIVERSITY *of* FLORIDA

LEADING THE CHARGE, CHARGING AHEAD